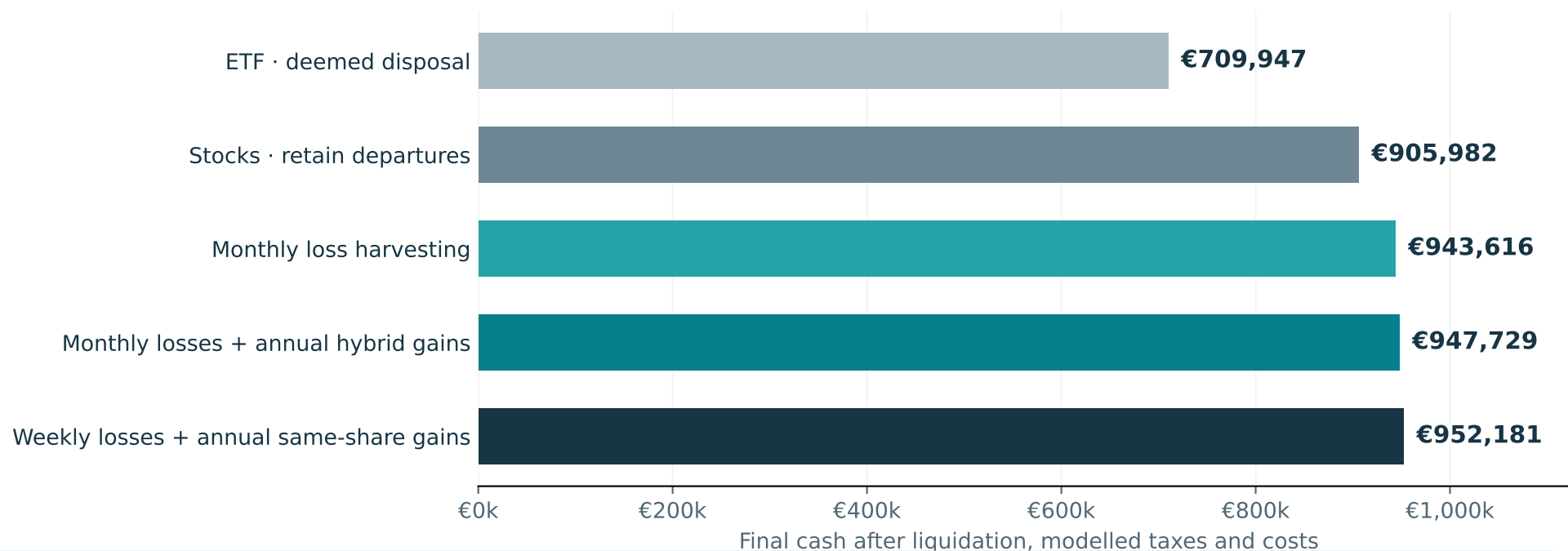


Nasdaq-100, after Irish tax

€1,000 each month · 16 years · €192,000 contributed · Final liquidation



Scenario	Total modelled tax	Trading costs	Wealth vs stock baseline
ETF	€317,451	€0	€-196,035
Stock baseline	€361,516	€2,236	€+0
Monthly losses	€382,199	€2,929	€+37,634
Monthly + hybrid gains	€374,397	€3,346	€+41,747
Weekly + same-share gains	€377,472	€3,064	€+46,198

Highest selected historical result: Weekly losses + annual same-share gains (€952,181). The stock baseline retains departed holdings and disables discretionary loss and annual gain harvesting. Differences include changed holdings and returns, not only tax savings.

Measure each change against its own control

Effects measured against different controls should not be added together

Feature / existing rules	After-tax wealth change	CGT reduction	Extra nominal exemption relief	Extra trading costs
Departure management No voluntary harvesting	€+26,600	€-13,001	€+166	€+211
Loss harvesting Monthly; annual gains off	€+11,034	€-8,083	€-1,753	€+482
Annual gain harvesting Monthly losses; departed first	€+4,113	€+7,206	€+6,364	€+417
Loss harvesting Weekly; annual gains off	€+9,122	€-7,211	€-1,753	€+358
Annual gain harvesting Weekly losses; restore shares	€+10,476	€+3,731	€+5,948	€+260
Annual exemption Weekly combined strategy	€+28,844	€-4,537	€+6,368	€+252
Annual exemption Monthly hybrid strategy	€+764	€+9,406	€+6,785	€+376

Wealth change compares complete replays. CGT reduction is the difference in modelled CGT paid; changed investments can also change taxable returns. Nominal exemption relief is 33% of the difference in exempt gains on the two ledgers. Neither number is a separate bonus to add to final wealth.

Exemption sensitivities switch the allowance off/on; gain-harvesting comparisons change trading rules. In the weekly case, the allowance changes full-strategy wealth by €28,844, but nominal exemption relief is €6,368. Tax capacity changes later holdings and cash decisions: the wealth change is not an allowance tax saving. Four interaction-control pairs remain in `marginal_effects.csv`.

Interaction between loss and gain harvesting: Weekly losses / annual gains: €+6,960; Monthly losses / annual hybrid gains: €+2,004. These measure how the loss-harvesting effect changes when gain harvesting is enabled. Standalone effects cannot simply be combined; differences along a consistent sequence of controls do sum correctly.

The rules, and the relief actually used

All three selected stock strategies review departed holdings quarterly

Selected strategy	Loss and gain review schedule
Monthly loss harvesting	Monthly losses; no annual gain review.
Monthly + hybrid gains	Monthly losses; December gains sell departed holdings first, then restore current shares.
Weekly + same-share gains	Friday loss reviews; December gains sell and repurchase eligible current shares.

Strategy	Exempt gains before final year	Earlier years using full allowance	Nominal relief including final year	Net losses from TLH disposals
Monthly losses	€5	0	€421	€25,464
Monthly + hybrid gains	€19,291	14	€6,785	€30,771
Weekly + same-share gains	€18,028	14	€6,368	€29,008

Loss review: sell an eligible whole holding when its euro loss after fees reaches both 5% and €25. Require no same-class purchase in the preceding 29 days and block repurchase for the next 29 days. Reinvest into eligible current underweights; monthly contributions remain unchanged.

Departure review: sell confirmed former constituents only when existing realised losses, carried losses and the remaining exemption cover the gain without increasing CGT at that review. A profitable departure can therefore remain in the portfolio.

Annual gain review: partial FIFO sales use losses before the €1,270 allowance. Same-share mode ranks by gain per euro sold; hybrid ranks departures then current overweights. Immediate replacement requires non-losing selected lots, current membership and the purchase restriction; both legs pay costs. Hybrid departure sales fund current underweights.

Harvested losses are not permanent relief equal to loss × 33%: replacement shares can have lower cost basis and a larger later gain. This study touches 17 tax years, including partial 2010 and final-liquidation 2026. All figures are hypothetical; no unrelated gains consume the exemption.

Evidence, assumptions and practical limits

A corrected cached-data replay, not a forecast or a live broker execution test

PERIOD AND FRESHNESS • 30 September 2010 to 30 September 2026. Stock price inputs extend through the final session; the issuer ETF and FX source cache contains observations through 2 October 2026. The last usable weights are dated 31 August 2026 and available 7 September. September weights were not available until 7 October and cannot inform September trades.

SELECTION • The three selected strategies were chosen retrospectively from the previous 25-case exploration. Individual trades use contemporaneously available inputs and fixed review rules, but selection was not validated out of sample. This focused replay includes diagnostic controls and exemption-off sensitivities to explain mechanisms, not to establish a universally best strategy.

FROZEN TAX SCENARIO • 33% share CGT; €1,270 annual allowance; 52.35% marginal dividend tax; 38% ETF fund tax and eight-year deemed disposal. Losses precede the allowance; unused allowance expires. Dividend withholding credits are included. Rates are comparison assumptions, not each historical year's actual law or any account holder's personal tax status.

COSTS AND EXECUTION • Non-euro share trades incur 0.15% FX per side. ETF NAV already reflects fund expenses and fund-level withholding. The model does not certify historical spreads, broker availability or fills. CGT is reserved as it accrues and settled annually; no claimed benefit depends on statutory payment timing.

DATA LIMITS • Historical membership, weight availability, delisted-stock dividends and corporate actions remain provisional. Some unquoted successors use explicit nontradable marks. Foreign reorganisations and compulsory same-class receipts can complicate share matching. The ledger audit records material restrictions and arithmetic checks; it is not a tax ruling.

REPRODUCE • Run `python historical/run_tax_optimization_study.py --workers 3`, or render saved results with `python historical/run_tax_optimization_report.py`. The current comparison, marginal effects, tax relief, interaction table, ledgers and input/code hashes are all in `historical/results/latest/`. Source and checks are retained separately; earlier generated report archives are not inputs to this report.

Read the assumptions before interpreting small differences as an investable advantage.

Research only, not tax, legal or investment advice. See the README legal notice.

[Revenue](#) · [CGT calculation and exemption](#)

[Revenue](#) · [Share matching restrictions](#)

What underperformed — and why

Three weak active variants, rerun on current inputs against the appropriate matched controls

Control A (€943,304): monthly loss harvesting at 5% and €25, quarterly tax-budget departure reviews and annual same-share gain harvesting. Control B (€936,099): the same departure and gain rules, without discretionary loss harvesting. Results include final liquidation, taxes and costs.

Change / matched comparator	Final cash	Wealth change vs control	Extra trading costs	CGT reduction vs control
Harvest any monthly loss $\geq$ €1 Control A	€909,433	€-33,870	€+736	€+15,906
Retain departures; gains only Control B	€910,238	€-25,861	€-226	€+12,883
Review gains every month Control A	€912,876	€-30,428	€+78	€+15,048

HARVEST ANY MONTHLY LOSS $\geq$ €1 • Relaxing the loss threshold triggered many more sales and changed replacement holdings and purchase restrictions. Recorded fees rose, but explain only part of the wealth shortfall; harvesting more losses did not improve the eventual after-tax outcome.

RETAIN DEPARTURES; GAINS ONLY • Retaining departed stocks prevented tax-budget exits and reinvestment into current constituents. This portfolio paid less tax and fewer fees but produced less final wealth. Annual gain harvesting still improved the plain retained-stock baseline; retention underperformed the matched exit policy. It still beats the plain stock baseline by €4,256.

REVIEW GAINS EVERY MONTH • More frequent gain reviews triggered more winning disposals without materially increasing annual exemption use. Earlier basis resets also changed later loss eligibility and holdings. Lower taxes did not offset the weaker investment outcome; an annual allowance is not multiplied by monthly reviews.

Ledger evidence	Loss sales variant / control	Gain sales variant / control	Net TLH losses realised	Extra nominal exemption relief
Harvest any monthly loss $\geq$ €1	1,408 / 411	269 / 199	€34,887	€-120
Retain departures; gains only	0 / 0	143 / 139	€0	€-2
Review gains every month	426 / 411	717 / 199	€32,195	€-80

Retrospectively selected historical outcomes, not universal failures. Fees, holdings, tax basis and later decisions all change; the comparisons do not separately attribute each mechanism. Lower tax can also reflect lower investment gains.